

Huawei Solar Inverter Interface Specification

Professional Register Mapping & Industrial Control Plane Architecture Definitions (Modbus-RTU / Modbus-TCP Protocols)

1. Core Telemetry & Electrical Metrics (Real-Time Performance)

Signal Name	Addresses	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Active power	32080	I32	2	1000	kW	RO	Real-time active output power tracking loop.
Reactive power	32082	I32	2	1000	kVar	RO	Real-time reactive power loops.
Peak active power of current day	32078	I32	2	1000	kW	RO	Highest daily active generation capacity peak achieved.
Power factor	32084	I16	1	1000	N/A	RO	Inverter system power factor scaling quotient.
Efficiency	32086	U16	1	100	%	RO	Real-time DC-to-AC conversion efficacy calculation output.

2. Safety Diagnostics & Thermal Management

Signal Name	Addresses	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Device status	32089	U16	1	1	N/A	RO	0x0000: Standby: initializing 0x0001: Standby: detecting insulation resistance 0x0002: Standby: detecting irradiation 0x0003: Standby: grid detecting 0x0100: Starting 0x0200: On-grid 0x0201: Grid connection: power limited 0x0202: Grid connection: self-derating 0x0203: Off-grid running 0x0300: Shutdown: fault 0x0301: Shutdown: command 0x0302: Shutdown: OVGR 0x0303: Shutdown: communication disconnected 0x0304: Shutdown: power limited 0x0305: Shutdown: manual startup required 0x0306: Shutdown: DC switches disconnected 0x0307: Shutdown: rapid cutoff 0x0308: Shutdown: input underpower 0x0401: Grid scheduling: cosφ-P curve 0x0402: Grid scheduling: Q-U curve 0x0403: Grid scheduling: PF-U curve 0x0404: Grid scheduling: dry contact 0x0405: Grid scheduling: Q-P curve 0x0500: Spot-check ready 0x0501: Spot-checking 0x0600: Inspecting 0x0700: AFCI self check 0x0800: I-V scanning 0x0900: DC input detection 0x0A00: Running: off-grid charging 0xA000: Standby: no irradiation
Fault code	32090	U16	1	1	N/A	RO	Central diagnostic identifier mapping to unique device fault registers.
Internal temperature	32087	I16	1	10	°C	RO	Inverter system cabinet thermal ambient core monitoring loop.
Insulation resistance	32088	U16	1	1000	MΩ	RO	DC array ground fault barrier check metric. Safety threshold indicator.

3. System States & Multi-Bit Indicators (Status Bitfields)

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
State 1	32000	Bitfield16	1	1	N/A	RO	Bit 0: Standby Bit 1: Grid-connected Bit 2: Grid-connected normally Bit 3: Grid connection with derating due to power rationing Bit 4: Grid connection with derating due to internal inverter causes Bit 5: Normal stop Bit 6: Stop due to faults Bit 7: Stop due to power rationing Bit 8: Shutdown Bit 9: Spot check
State 2	32002	Bitfield16	1	1	N/A	RO	Bit 0: Locking status (0: locked; 1: unlocked) Bit 1: PV connection status (0: disconnected; 1: connected) Bit 2: DSP data collection (0: no; 1: yes)
State 3	32003	Bitfield32	2	1	N/A	RO	Bit 0: Off-grid (0: on-grid; 1: off-grid) Bit 1: Off-grid switch (0: disable; 1: enable)
Alarm 1	32008	Bitfield16	1	1	N/A	RO	Bit 0: High String Input Voltage (ID: 2001) Bit 1: DC Arc Fault (ID: 2002) Bit 2: String Reverse Connection (ID: 2011) Bit 3: String Current Backfeed (ID: 2012) Bit 4: Abnormal String Power (ID: 2013) Bit 5: AFCI Self-Check Fail (ID: 2021) Bit 6: Phase Wire Short to PE (ID: 2031) Bit 7: Grid Loss (ID: 2032) Bit 8: Grid Undervoltage (ID: 2033) Bit 9: Grid Overvoltage (ID: 2034) Bit 10: Grid Volt. Imbalance (ID: 2035)
Alarm 2	32009	Bitfield16	1	1	N/A	RO	Bit 0: Abnormal Residual Current (ID: 2051) Bit 1: Abnormal Grounding (ID: 2061) Bit 2: Low Insulation Resistance (ID: 2062) Bit 3: Overtemperature (ID: 2063) Bit 4: Device Fault (ID: 2064) Bit 5: Upgrade Failed/Version Mismatch (ID: 2065) Bit 6: License Expired (ID: 2066) Bit 7: Faulty Monitoring Unit (ID: 61440) Bit 8: Faulty Power Collector (ID: 2067) Bit 9: Battery Abnormal (ID: 2068) Bit 10: Active Islanding (ID: 2070)
Alarm 3	32010	Bitfield16	1	1	N/A	RO	Bit 0: Optimizer Fault (ID: 2081) Bit 1: Built-in PID Operation Abnormal (ID: 2085) Bit 2: High Input String Voltage to Ground (ID: 2014) Bit 3: External Fan Abnormal (ID: 2086) Bit 4: Battery Reverse Connection (ID: 2069) Bit 5: On/Off Grid Ctrl Abnormal (ID: 2082) Bit 6: PV String Loss (ID: 2015) Bit 7: Internal Fan Abnormal (ID: 2087) Bit 8: DC Protection Unit Abnormal (ID: 2088) Bit 9: EL Unit Abnormal (ID: 2089) Bit 10: Active Adjustment Instruction Fault (ID: 2090)

4. Energy Accumulation & Yield Statistics

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
daily energy yield	32114	U32	2	100	kWh	RO	Yield generation accumulated within the current daily tracking loop.
accumulated energy yield	32106	U32	2	100	kWh	RO	Lifetime absolute continuous active yield tracking summation standard registry.

5. System Temporal Context & Synchronization

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
System time	40000	U32	2	1	N/A	RW	Central clock context synchronization register.
Startup time	32091	U32	2	1	N/A	RO	Daily execution dawn initialization timestamp context (Epoch seconds).
Shutdown time	32093	U32	2	1	N/A	RO	Closing operational threshold suspension execution timestamp context (Epoch seconds).

6. Remote Control Commands & Execution Configuration

Signal Name	Address	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Startup	40200	U16	1	1	N/A	WO	Write 1 to force manual absolute programmatic hardware runtime startup sequence trigger.
Shutdown	40201	U16	1	1	N/A	WO	Write 1 to force manual absolute programmatic hardware operational shutdown trigger.

7. Power Scheduling & Grid Compliance Parameters

Signal Name	Addresses	Type	Qty	Gain	Unit	R/W	Operational Scope & Bitfield Decodings
Failsafe Active Power Limit [kw][High Accuracy]	42405	I32	2	1000	kW	RW	Power safety limit boundary baseline enforcement. Range context: [0.000, Pmax].
Fast power scheduling	45086	U16	1	1	N/A	RW	Enables or disables rapid grid telemetry scaling protocol (0: Disable 1: Enable).
Grid code	42000	U16	1	1	N/A	RW	National compliance safety boundary selector. Maps to Table 5-3 region maps (e.g., 0: VDE-AR-N-4105 Germany, 1: NB/T 32004 China, 2: UTE France, 13: CEI0-21 Italy, 19: AS4777 Australia).